

Can ChatGPT suggest patterns? An exploratory study about answers given by AI-assisted tools to design problems.

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Abstract. General-purpose AI-assisted tools, such as ChatGPT, have recently gained much attention from the media and the general public. That raised questions about in which tasks we can apply such a tool. A good code design is essential for agile software development to keep it ready for change. In this context, identifying which design pattern can be appropriate for a given scenario can be considered an advanced skill that requires a high degree of abstraction and a good knowledge of object orientation. This paper aims to perform an exploratory study investigating the effectiveness of an AI-assisted tool in assisting developers in choosing a design pattern to solve design scenarios. To reach this goal, we gathered 56 existing questions used by teachers and public tenders that provide a concrete context and ask which design pattern would be suitable. We submitted these questions to ChatGPT and analyzed the answers. We found that 93% of the questions were answered correctly with a good level of detail, demonstrating the potential of such a tool as a valuable resource to help developers to apply design patterns and make design decisions.

Keywords: ChatGPT, Design Patterns, Artificial Intelligence, AI-assisted tools